

Electric Field Due to a Uniformly Charged Plane (Ring Method)

Setup. Consider a uniformly charged plane (sheet) in the $z = 0$ plane with surface charge density

$$\sigma \text{ (C/m}^2\text{)}.$$

Find the electric field at the observation point

$$P(0, 0, z), \quad z > 0.$$

Idea: choose a ring on the sheet

Consider a thin ring on the sheet centered on the z -axis (directly below P), of radius r and thickness dr .

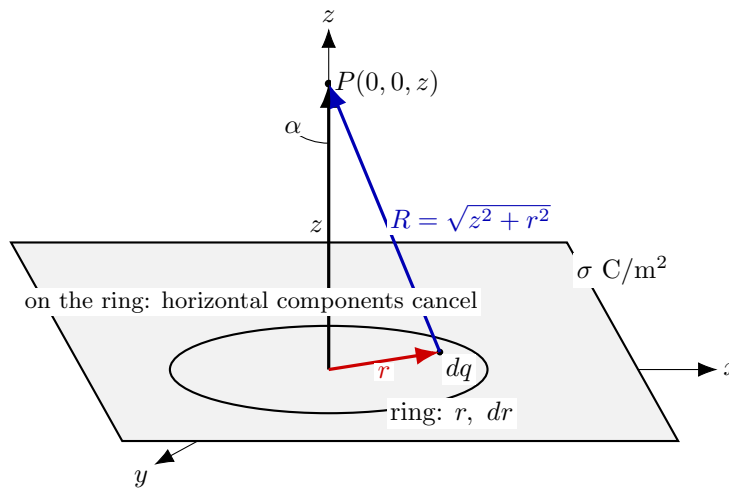


Figure 1: Ring element method for a uniformly charged plane. Points on the ring are equidistant from P , so in-plane components cancel and only the vertical component survives.

Charge on the ring

Area of the thin ring is

$$dA = (2\pi r) dr,$$

hence the charge in the ring is

$$dq = \sigma dA = \sigma(2\pi r dr).$$

Field contribution from the ring

Every point on the ring is at the same distance

$$R = \sqrt{z^2 + r^2}$$



from P . Due to symmetry, for every field contribution there is an equal and opposite in-plane component from the opposite side of the ring. Therefore, **only the vertical (z) component survives**.

Magnitude due to dq is

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{R^2} = \frac{1}{4\pi\epsilon_0} \frac{dq}{z^2 + r^2}.$$

The vertical component is

$$dE_z = dE \cos \alpha, \quad \cos \alpha = \frac{z}{R} = \frac{z}{\sqrt{z^2 + r^2}}.$$

Hence,

$$dE_z = \frac{1}{4\pi\epsilon_0} \frac{dq}{z^2 + r^2} \cdot \frac{z}{\sqrt{z^2 + r^2}} = \frac{1}{4\pi\epsilon_0} dq \frac{z}{(z^2 + r^2)^{3/2}}.$$

Substituting $dq = \sigma(2\pi r dr)$,

$$\boxed{dE_z = \frac{\sigma}{2\epsilon_0} \frac{z r}{(z^2 + r^2)^{3/2}} dr} \quad (\text{direction: } +\hat{\mathbf{z}} \text{ for } z > 0, \sigma > 0).$$

Result (disk of radius a , then infinite sheet as a limit)

If the charged plane is a disk of radius a , integrate r from 0 to a :

$$E_z(z) = \int_0^a dE_z = \frac{\sigma}{2\epsilon_0} \int_0^a \frac{z r}{(z^2 + r^2)^{3/2}} dr = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + a^2}} \right).$$

Integration steps (by substitution)

Starting from

$$E_z(z) = \frac{\sigma}{2\epsilon_0} \int_0^a \frac{z r}{(z^2 + r^2)^{3/2}} dr,$$

use the substitution

$$u = z^2 + r^2 \Rightarrow du = 2r dr \Rightarrow r dr = \frac{1}{2} du.$$

Then

$$E_z(z) = \frac{\sigma}{2\epsilon_0} \int_{u=z^2}^{u=z^2+a^2} \frac{z}{u^{3/2}} \left(\frac{1}{2} du \right) = \frac{\sigma}{2\epsilon_0} \cdot \frac{z}{2} \int_{z^2}^{z^2+a^2} u^{-3/2} du.$$

Now integrate:

$$\int u^{-3/2} du = \frac{u^{-1/2}}{-1/2} = -2u^{-1/2}.$$

Therefore,

$$E_z(z) = \frac{\sigma}{2\epsilon_0} \cdot \frac{z}{2} \left[-2u^{-1/2} \right]_{z^2}^{z^2+a^2} = \frac{\sigma}{2\epsilon_0} z \left[\frac{1}{\sqrt{z^2}} - \frac{1}{\sqrt{z^2 + a^2}} \right].$$

For $z > 0$, $\sqrt{z^2} = z$, hence

$$E_z(z) = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + a^2}} \right].$$



$$\mathbf{E}(z) = E_z(z) \hat{\mathbf{z}} = \frac{\sigma}{2\varepsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + a^2}} \right) \hat{\mathbf{z}}$$

For an **infinite sheet**, take $a \rightarrow \infty$:

$$\mathbf{E} = \frac{\sigma}{2\varepsilon_0} \hat{\mathbf{z}} \quad (z > 0), \quad \mathbf{E} = -\frac{\sigma}{2\varepsilon_0} \hat{\mathbf{z}} \quad (z < 0).$$

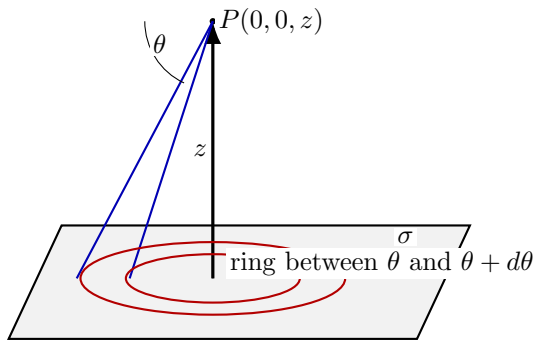
Key Takeaways

- Use a ring element: $dq = \sigma(2\pi r dr)$ and $R = \sqrt{z^2 + r^2}$.
- By symmetry, in-plane components cancel; only the z -component survives.
- Disk of radius a :

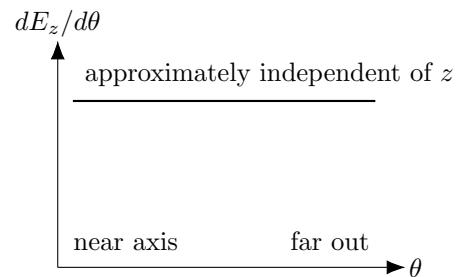
$$\mathbf{E}(z) = \frac{\sigma}{2\varepsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + a^2}} \right) \hat{\mathbf{z}} \quad (z > 0).$$

- Infinite sheet limit ($a \rightarrow \infty$): $\mathbf{E} = \frac{\sigma}{2\varepsilon_0} \hat{\mathbf{z}}$ above the sheet and opposite below.

Intuitive explanation: why E becomes nearly constant for a large plane sheet



A narrow cone from P intercepts a ring on the plane.



Contribution per angular slice (qualitative).

Figure 2: Cone-to-ring picture: when the sheet is very large, widening ring area competes with increasing distance so the net field tends to a constant.

Consider a uniformly charged plane (surface charge density σ) in the $z = 0$ plane. Let the observation point be $P(0, 0, z)$ with $z > 0$.

Key geometric picture: a fixed viewing cone intercepts a ring on the plane

Look from P towards the plane. A narrow cone of directions between angles θ and $\theta + d\theta$ intersects the plane in a thin *ring*. The crucial point is:

- As z increases, the ring moves outward to larger radius r and becomes larger.
- At the same time, the distance to the ring also increases, making each small charge element weaker.
- For an *infinite (or very large) plane*, these effects balance so that the total contribution from an angular slice is essentially independent of z .

Ring intercepted by a cone (relation between r and θ)

For a ray making angle θ with the z -axis:

$$r = z \tan \theta.$$

A small change $d\theta$ produces a change

$$dr = z \sec^2 \theta d\theta.$$

The corresponding ring area is

$$dA = (2\pi r) dr = 2\pi(z \tan \theta)(z \sec^2 \theta d\theta) = 2\pi z^2 \tan \theta \sec^2 \theta d\theta.$$

Hence the charge in that ring is

$$dq = \sigma dA = 2\pi\sigma z^2 \tan \theta \sec^2 \theta d\theta.$$

Why the contribution does not die off with z (balancing argument)

Take the field contribution from the ring at angle θ .

- Distance from ring to P is
- $$R = \sqrt{z^2 + r^2} = z \sec \theta.$$
- Field magnitude from dq scales as dq/R^2 .
 - Only the vertical component survives by symmetry; multiply by $\cos \theta$.

So, up to the constant factor $1/(4\pi\epsilon_0)$,

$$dE_z \propto \frac{dq}{R^2} \cos \theta.$$

Substitute dq and R :

$$\frac{dq}{R^2} \cos \theta = \left(2\pi\sigma z^2 \tan \theta \sec^2 \theta d\theta\right) \cdot \frac{1}{z^2 \sec^2 \theta} \cdot \cos \theta = 2\pi\sigma \tan \theta \cos \theta d\theta = 2\pi\sigma \sin \theta d\theta.$$

Notice: the z^2 cancels completely. That is the intuitive reason the field does *not* decay with z for an (approximately) infinite plane: as z increases, the ring you see at a given θ contains more charge ($\propto z^2$), but it is also farther away ($R^2 \propto z^2$), producing a balance.



Constant-field limit for a very large (infinite) plane

Including the constant factor,

$$dE_z = \frac{1}{4\pi\epsilon_0} (2\pi\sigma \sin\theta d\theta).$$

For an infinite plane, the cone extends to $\theta \rightarrow \pi/2$:

$$E_z = \int_0^{\pi/2} dE_z = \frac{1}{4\pi\epsilon_0} 2\pi\sigma \int_0^{\pi/2} \sin\theta d\theta = \frac{1}{4\pi\epsilon_0} 2\pi\sigma [1] = \boxed{\frac{\sigma}{2\epsilon_0}}.$$

Direction:

$$\mathbf{E} = +\frac{\sigma}{2\epsilon_0} \hat{\mathbf{z}} \quad (z > 0), \quad \mathbf{E} = -\frac{\sigma}{2\epsilon_0} \hat{\mathbf{z}} \quad (z < 0).$$

Connection to the finite disk result

For a disk of radius a , the largest visible angle is $\theta_{\max} = \tan^{-1}(a/z)$. As $a \gg z$, $\theta_{\max} \rightarrow \pi/2$, so the field approaches the constant infinite-sheet value.

Key Takeaways

- A narrow cone of directions from P intersects the plane in a ring; as z increases, ring area grows.
- Increased charge in the ring (more area) balances increased distance, yielding a net contribution that is nearly z -independent.
- For a sufficiently large plane ($a \gg z$), the field approaches the infinite-sheet constant value.

Example: Force Between Two Parallel (Infinite) Line Charges

Setup. Two infinitely long, parallel line charges are separated by distance d . Let line (1) carry linear charge density λ_1 (C/m) and line (2) carry linear charge density λ_2 (C/m).

Field due to line (1) at the location of line (2)

From the previous result for an infinite line charge, the electric field magnitude at perpendicular distance d from line (1) is

$$E_1(d) = \frac{\lambda_1}{2\pi\epsilon_0 d}$$

and the direction is radially outward from line (1) (for $\lambda_1 > 0$), inward (for $\lambda_1 < 0$).

Force per unit length on line (2)

A segment of line (2) of length L carries charge $Q_2 = \lambda_2 L$. Hence the force on that segment is

$$F = Q_2 E_1(d) = \lambda_2 L E_1(d).$$

Therefore, the **force per unit length** (often denoted f) is

$$\boxed{\frac{F}{L} = \lambda_2 E_1(d) = \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0 d}}$$

Direction:

- If $\lambda_1 \lambda_2 > 0$ (same sign), the force is **repulsive**.
- If $\lambda_1 \lambda_2 < 0$ (opposite sign), the force is **attractive**.

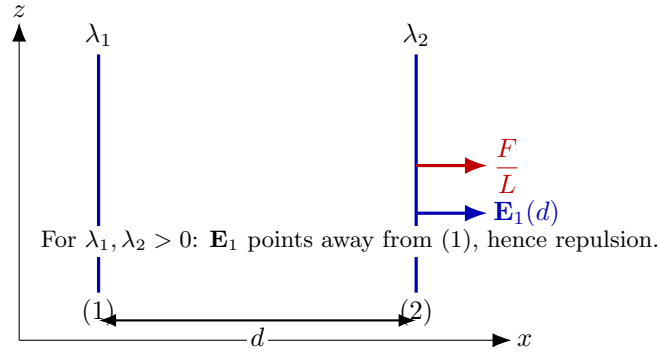


Figure 3: Two parallel infinite line charges separated by d . Field of line (1) at line (2) gives force per unit length on line (2).

Interpretation and practical remark

The expression

$$\frac{F}{L} = \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0 d}$$

suggests the total force becomes unbounded as $L \rightarrow \infty$ because the total charge also grows without limit. Therefore, for infinite lines we report the physically meaningful quantity **force per unit length**.

Practical relevance. Perfectly infinite line charges do not exist, but long conductors (e.g., overhead power lines) can be approximated as line charges over distances d much smaller than the line length. For AC power lines (typically 50–60 Hz), the field can often be treated as *quasi-static* at distances small compared to the electromagnetic wavelength.

Key Takeaways

- Field of line (1) at distance d : $E_1(d) = \frac{\lambda_1}{2\pi\epsilon_0 d}$.

- Force per unit length on line (2):

$$\frac{F}{L} = \lambda_2 E_1(d) = \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0 d}.$$

- Same-sign line charges repel; opposite-sign attract.
- Infinite lines are idealizations; F/L is the physically meaningful quantity.

Why Coulomb's Force/Field Integral is Often Not Sufficient

Coulomb's law in integral form (needs the full charge distribution)

In electrostatics, the electric field at an observation point $\mathbf{r}$ due to a (volume) charge distribution $\rho(\mathbf{r}')$ is written as

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint_V \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{\|\mathbf{r} - \mathbf{r}'\|^3} d\tau'$$

where $\mathbf{r}$ is the observation point and $\mathbf{r}'$ is the source point.

Critical requirement. To use the above equation directly, we must know the charge density $\rho(\mathbf{r}')$ *everywhere* in space (or at least everywhere in the region V where charge resides).

Why this is a problem in practice

Usually, we do *not* know where the charge is exactly:

- In conductors, charges are free to move and rearrange themselves until electrostatic equilibrium is reached.
- In many engineering structures, charge appears on surfaces and redistributes in a way that depends on geometry and nearby conductors/dielectrics.
- Even if the total charge is known, the *spatial distribution* (surface/volume density) is generally unknown a priori.

Example: A charged metallic ball (unknown surface distribution)

Consider a metallic (conducting) sphere, and suppose we place a net charge of 5 C on it. Because it is a conductor, charges move and settle on the surface in a configuration that depends on the geometry and the surrounding environment.



In such cases:

- We may know the *total* charge ($Q = 5 \text{ C}$),
- We know the *shape* of the conductor (sphere),
- But we may *not* know the exact surface charge density $\sigma(\mathbf{r}')$ everywhere on the surface, especially when other conductors/dielectrics are nearby.

Consequence: Coulomb's integral cannot be used directly unless the charge distribution is already known.

Engineering motivation: capacitance and field solutions

A very important component in electrical engineering is the **capacitor**: two conductors (often plates) separated by a distance with a dielectric in between. The key questions are:

- What is the electric field between the conductors?
- How is charge distributed on each conductor surface?
- What is the potential difference V for a given charge Q (and hence the capacitance $C = Q/V$)?

Key Takeaways

- Coulomb's integral requires the full charge distribution (ρ or σ) everywhere, which is often unknown a priori.
- In conductors, charges redistribute depending on geometry and surroundings; you may know total Q but not $\sigma(\mathbf{r})$.
- Practical problems (e.g., capacitors) are typically solved via boundary-value methods (Laplace/Poisson + boundary conditions), then σ is obtained from the field.: We will cover these aspects in coming lectures

Electrostatic Potential: Work-Energy View (Analogy with Gravity)

1) Warm-up: work and potential energy in a gravitational field

In mechanics, the infinitesimal work done by a force $\mathbf{F}$ along a small displacement $d\mathbf{l}$ is

$$dW = \mathbf{F} \cdot d\mathbf{l}.$$

Near the Earth, the gravitational force on a mass m is approximately constant:

$$\mathbf{F}_g = m\mathbf{g}, \quad \mathbf{g} = -g\hat{\mathbf{z}}.$$

If the mass moves from height z_A to z_B , the work done by gravity is

$$W_g(A \rightarrow B) = \int_A^B \mathbf{F}_g \cdot d\mathbf{l} = \int_A^B (-mg \hat{\mathbf{z}}) \cdot (dz \hat{\mathbf{z}}) = -mg \int_{z_A}^{z_B} dz = mg(z_A - z_B).$$

Hence, when the mass moves *down* by height h (i.e., $z_A - z_B = h$),

$$W_g = +mgh \quad (\text{gravity does positive work and potential energy decreases}).$$

This motivates the gravitational potential energy

$$U_g(z) = mgz, \quad \Delta U_g = U_g(B) - U_g(A) = mg(z_B - z_A),$$

so that

$$W_g(A \rightarrow B) = -\Delta U_g.$$

Potential energy decreases: $\Delta U_g = -mgh$.

Gravity does positive work for downward motion: $W_g = +mgh$.

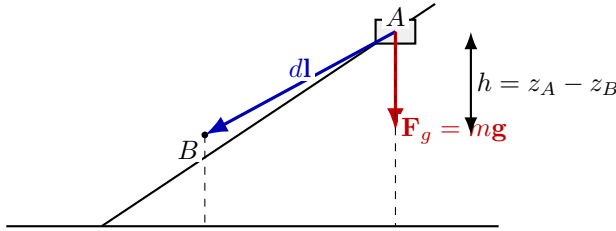


Figure 4: Correct height definition: the height change h is the *vertical* drop $z_A - z_B$, not the distance along the incline.

Work done in moving a charge along an arbitrary path

Suppose a test charge q is moved from point A to point B along a path C in the presence of an electric field $\mathbf{E}(\mathbf{r})$ (for example, the field produced by a source charge Q).

Step 1: Infinitesimal work

The electric force on q is

$$\mathbf{F}_e(\mathbf{r}) = q \mathbf{E}(\mathbf{r}).$$

The infinitesimal work done by this force over a small displacement $d\mathbf{l}$ is

$$dW = \mathbf{F}_e \cdot d\mathbf{l} = q \mathbf{E}(\mathbf{r}) \cdot d\mathbf{l}.$$

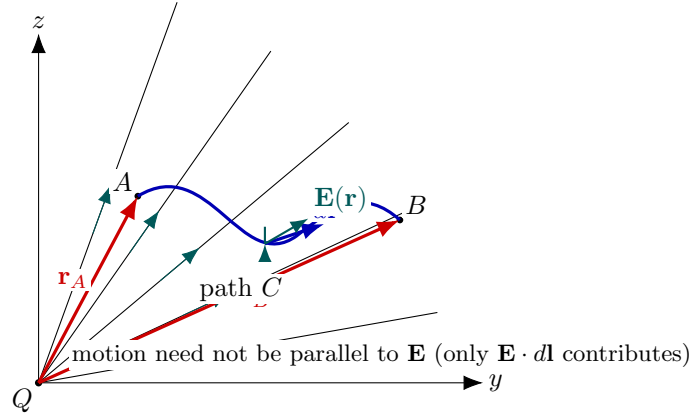


Figure 5: A charge q moved from A to B along a path C in an electric field $\mathbf{E}(\mathbf{r})$. The work depends on the component of $\mathbf{E}$ along the tangent direction $d\mathbf{l}$.

Step 2: Approximate the path by small straight segments

Divide the curve C into N small segments with points

$$\mathbf{r}_0 = \mathbf{r}_A, \mathbf{r}_1, \dots, \mathbf{r}_N = \mathbf{r}_B, \quad \Delta \mathbf{r}_k = \mathbf{r}_{k+1} - \mathbf{r}_k.$$

Over a very small segment, take the field as approximately constant:

$$\mathbf{E}(\mathbf{r}) \approx \mathbf{E}(\mathbf{r}_k) \quad \text{on segment } k.$$

Then the work over segment k is approximately

$$\Delta W_k \approx q \mathbf{E}(\mathbf{r}_k) \cdot \Delta \mathbf{r}_k.$$

Summing over all segments,

$$W_C(A \rightarrow B) \approx q \sum_{k=0}^{N-1} \mathbf{E}(\mathbf{r}_k) \cdot \Delta \mathbf{r}_k.$$

Step 3: Take the limit (line integral)

As the segmentation is refined ($\max \|\Delta \mathbf{r}_k\| \rightarrow 0$), the Riemann sum becomes a line integral:

$$W_C(A \rightarrow B) = q \int_C \mathbf{E}(\mathbf{r}) \cdot d\mathbf{l} \quad (\text{path/line integral}).$$

Optional detail: tangent form

If the curve is parametrized by arc length s , then

$$d\mathbf{l} = \hat{\mathbf{t}} ds,$$

where $\hat{\mathbf{t}}$ is the unit tangent to the path. Hence,

$$W_C = q \int_C \mathbf{E} \cdot \hat{\mathbf{t}} ds.$$

Only the component of $\mathbf{E}$ along the direction of motion contributes to the work.

Path Independence of Work in the Coulomb Field (Point Charge)

Setup. A source charge Q is fixed at the origin. A test charge q is moved from point A to point B along an arbitrary path C (not necessarily radial). Let

$$r_A = \|\mathbf{r}_A\|, \quad r_B = \|\mathbf{r}_B\|$$

be the distances of A and B from the origin.

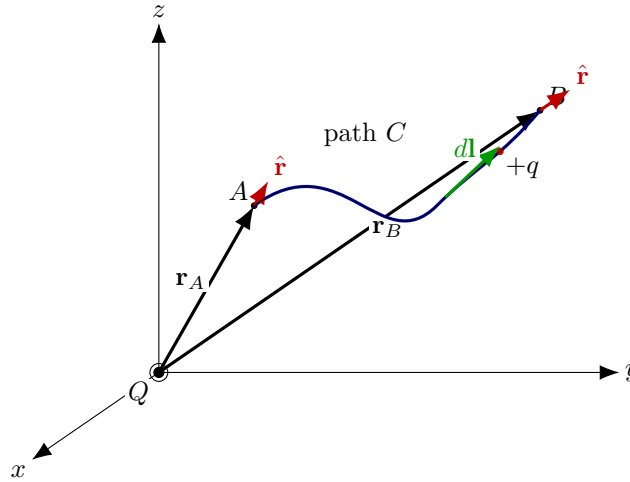


Figure 6: Source charge Q at the origin; a positive test charge $+q$ moves along path C . The radial unit vector $\hat{\mathbf{r}}$ is shown outward at A and B , and $d\mathbf{l}$ is drawn exactly tangent to the path.

Step 1: Coulomb force field

The Coulomb force on the test charge q at position $\mathbf{r}$ is

$$\mathbf{F}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}, \quad r = \|\mathbf{r}\|.$$

Step 2: Work as a line integral

The work done by the electrostatic force in moving q from A to B along path C is

$$W_C(A \rightarrow B) = \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{l}.$$

Step 3: The key simplification $\hat{\mathbf{r}} \cdot d\mathbf{l} = dr$

Write the infinitesimal displacement in spherical coordinates:

$$d\mathbf{l} = \hat{\mathbf{r}} dr + \hat{\boldsymbol{\theta}} r d\theta + \hat{\boldsymbol{\phi}} r \sin \theta d\phi.$$

Dot with $\hat{\mathbf{r}}$:

$$\hat{\mathbf{r}} \cdot d\mathbf{l} = \hat{\mathbf{r}} \cdot (\hat{\mathbf{r}} dr) + \hat{\mathbf{r}} \cdot (\hat{\boldsymbol{\theta}} r d\theta) + \hat{\mathbf{r}} \cdot (\hat{\boldsymbol{\phi}} r \sin \theta d\phi) = dr,$$

because $\hat{\mathbf{r}} \perp \hat{\boldsymbol{\theta}}$ and $\hat{\mathbf{r}} \perp \hat{\boldsymbol{\phi}}$.

Step 4: Evaluate the integral (path independent)

Substitute $\mathbf{F}$ into the work integral:

$$W_C = \int_C \left(\frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{l} = \frac{qQ}{4\pi\epsilon_0} \int_C \frac{1}{r^2} (\hat{\mathbf{r}} \cdot d\mathbf{l}).$$

Using $\hat{\mathbf{r}} \cdot d\mathbf{l} = dr$,

$$W_C = \frac{qQ}{4\pi\epsilon_0} \int_{r_A}^{r_B} \frac{1}{r^2} dr = \frac{qQ}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_{r_A}^{r_B}.$$

Therefore,

$$W_C(A \rightarrow B) = \frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right)$$

which depends only on the endpoints r_A and r_B and **not** on the path C .

Conclusion (path independence)

$$\int_C \mathbf{F} \cdot d\mathbf{l} \text{ is path independent for the Coulomb force field.}$$

Equivalently, since $\mathbf{F} = q\mathbf{E}$,

$$\int_C \mathbf{E} \cdot d\mathbf{l} = \frac{1}{q} \int_C \mathbf{F} \cdot d\mathbf{l}$$

is also path independent in electrostatics.

Potential Energy and Electrostatic Potential (from Path-Independent Work)

1) Work done by the electrostatic force (Coulomb field)

From the previous result for a source charge Q at the origin,

$$W_{A \rightarrow B} = \int_C \mathbf{F} \cdot d\mathbf{l} = \frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right) \quad (\text{independent of the path } C).$$



2) Work–energy relation and definition of potential energy

In electrostatics, the electric force is conservative. Therefore, the work done by the electric field is equal to *minus* the change in potential energy:

$$W_{A \rightarrow B} = -\Delta U = -(U_B - U_A).$$

Hence,

$$U_B - U_A = -W_{A \rightarrow B} = -\frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right) = \frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r_B} - \frac{1}{r_A} \right).$$

So the potential energy at B relative to A is

$$U_B = U_A + \frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r_B} - \frac{1}{r_A} \right).$$

3) Choosing the reference at infinity

A standard (and very useful) choice is to set the reference potential energy to zero at infinity:

$$U(\infty) = 0.$$

Take A to be “far away” ($r_A \rightarrow \infty$), and let B be a point at distance r from the origin. Then

$$U(r) - U(\infty) = \frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{\infty} \right) = \frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r} - 0 \right).$$

Therefore,

$$U(r) = \frac{qQ}{4\pi\epsilon_0 r}.$$

4) Definition of electrostatic potential (scalar)

We now define the **electrostatic potential** (often denoted ϕ or V) as

$$\phi(\mathbf{r}) = \frac{U(\mathbf{r})}{q} \quad (\text{potential energy per unit charge}).$$

For a point charge Q at the origin,

$$\phi(r) = \frac{Q}{4\pi\epsilon_0 r}.$$

This is a **scalar** field (simpler than the vector field $\mathbf{E}$).

5) Relation between potential and field (integral form)

Starting from $W = -\Delta U$ and $W = q \int_A^B \mathbf{E} \cdot d\mathbf{l}$, we obtain

$$q \int_A^B \mathbf{E} \cdot d\mathbf{l} = -(U_B - U_A).$$

Divide by q :

$$\int_A^B \mathbf{E} \cdot d\mathbf{l} = -(\phi_B - \phi_A).$$

Hence,

$$\boxed{\phi_B - \phi_A = - \int_A^B \mathbf{E} \cdot d\mathbf{l}} \quad (\text{path independent in electrostatics}).$$

With the reference $\phi(\infty) = 0$:

$$\boxed{\phi(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}.$$

Key Takeaways

- For a point charge, the work done by the electrostatic force is path independent:

$$W_{A \rightarrow B} = \frac{qQ}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right).$$

- Conservative field: $W_{A \rightarrow B} = -(U_B - U_A)$.
- With $U(\infty) = 0$, the potential energy of q at distance r is

$$U(r) = \frac{qQ}{4\pi\epsilon_0 r}.$$

- Electrostatic potential (scalar):

$$\phi(r) = \frac{U}{q} = \frac{Q}{4\pi\epsilon_0 r}.$$

- With the reference $\phi(\infty) = 0$:

$$\phi(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}.$$